3. Keyless Entry System

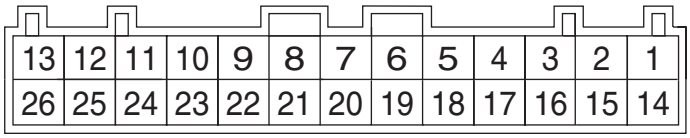
A: WIRING DIAGRAM

1. KEYLESS ENTRY

<Ref. to WI-168, WIRING DIAGRAM, Keyless Entry System.>

B: ELECTRICAL SPECIFICATION

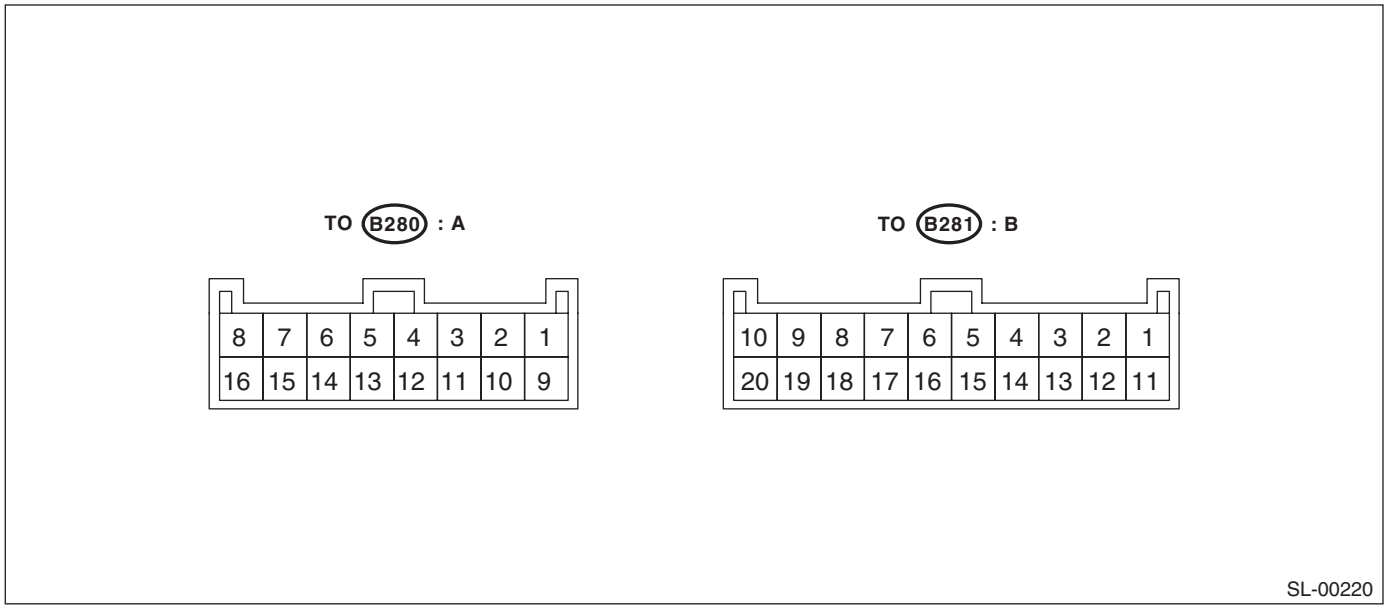
1. KEYLESS ENTRY CONTROL MODULE



SL-00478

Contents	Terminal No.	Measuring condition
Turn signal light LH	1 (OUTPUT)	Battery voltage is detected when the transmitter UNLOCK/DISARM button or the LOCK/ARM button is pressed.
Power supply (Backup)	2	Battery voltage is constantly present.
Keyless buzzer	3 (OUTPUT)	0 V is detected when the transmitter UNLOCK/DISARM button or the LOCK/ARM button is pressed.
Door unlock switch	4 (INPUT)	0 V is detected when the door lock switch is operated.
Door lock switch	5 (INPUT)	0 V is detected when the door lock switch is operated.
Door switch and rear gate latch switch	6 (INPUT)	0 V is detected when one of the doors or the rear gate is opened.
Trunk latch switch	7 (INPUT)	0 V is detected when the trunk is opened.
Key warning switch	9 (INPUT)	Battery voltage is present when the key is inserted into the ignition switch.
Ignition switch (ON)	10 (INPUT)	Battery voltage is present when the ignition switch is turned to ON.
Turn signal light RH	13 (OUTPUT)	Battery voltage is detected when the transmitter UNLOCK/DISARM button or the LOCK/ARM button is pressed.
Ground	14	0 V is constantly present.
Body integrated unit	18 (OUTPUT)	Battery voltage is detected when pressing the transmitter UNLOCK/DISARM button.
Body integrated unit	19 (OUTPUT)	Battery voltage is detected when pressing the transmitter LOCK/ARM button.
Horn relay	24 (OUTPUT)	When the transmitter LOCK/ARM button is pressed 3 times within 5 seconds, 0 V is detected.
Power supply (Backup)	26	Battery voltage is constantly present.

2. BODY INTEGRATED UNIT



Contents	Terminal No.	Measuring condition
Door switch (except for the driver's side)	B7 (INPUT)	0 V is detected when the door is opened (except for the driver's side door).
Door switch (driver's side)	B8 (INPUT)	0 V is detected when the driver's side door is opened.
Door unlock switch	B11 (INPUT)	0 V is detected when the door unlock switch is operated.
Door lock switch	B12 (INPUT)	0 V is detected when the door lock switch is operated.
Keyless entry control module	B13 (INPUT)	Battery voltage is detected when pressing the transmitter LOCK/ARM button.
Keyless entry control module	B14 (INPUT)	Battery voltage is detected when pressing the transmitter UNLOCK/DISARM button.
Ignition switch (ON)	B19 (INPUT)	Battery voltage is present when the ignition switch is turned to ON.
Key warning switch	B20 (INPUT)	Battery voltage is present when the key is inserted into the ignition switch.
Power supply	A2	Battery voltage is constantly present.
Ground	A4	0 V is constantly present.
Room light	A5 (OUTPUT)	0 V is detected when pressing the transmitter UNLOCK/DISARM button.
Door and rear gate lock actuator	A6 (OUTPUT)	Battery voltage is detected when pressing the transmitter LOCK/ARM button.
Door and rear gate lock actuator (except for the driver's side)	A7 (OUTPUT)	Battery voltage is detected when pressing the transmitter UNLOCK/DISARM button twice.
Door lock actuator (driver's side)	A8 (OUTPUT)	Battery voltage is detected when pressing the transmitter UNLOCK/DISARM button once.
Ground	A13	0 V is constantly present.

C: INSPECTION

1. SYMPTOM CHART

Symptom	Repair order	Reference
None of the functions of the keyless entry system operate.	1. Check the keyless transmitter battery and function.	<Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>
	2. Check the fuse.	<Ref. to SL-16, CHECK FUSE, INSPECTION, Keyless Entry System.>
	3. Check the keyless entry control module power supply and ground circuit.	<Ref. to SL-16, CHECK POWER SUPPLY AND GROUND CIRCUIT, INSPECTION, Keyless Entry System.>
	4. Replace the keyless entry control module.	<Ref. to SL-55, Keyless Entry Control Module.>
The keyless transmitter cannot be registered.	1. Check the keyless transmitter battery and function.	<Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>
	2. Check the ignition switch circuit.	<Ref. to SL-17, CHECK IGNITION SWITCH CIRCUIT, INSPECTION, Keyless Entry System.>
	3. Check the door switch.	<Ref. to SL-17, CHECK DOOR SWITCH, INSPECTION, Keyless Entry System.>
	4. Replace the keyless entry control module.	<Ref. to SL-55, Keyless Entry Control Module.>
Door lock or unlock does not operate. NOTE: If the door lock control system does not operate when using the door lock switch, check the door lock control system. <Ref. to SL-8, INSPECTION, Door Lock Control System.>	1. Check the keyless transmitter battery and function.	<Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>
	2. Check the key warning switch.	<Ref. to SL-18, CHECK KEY WARNING SWITCH, INSPECTION, Keyless Entry System.>
	3. Check the door switch.	<Ref. to SL-17, CHECK DOOR SWITCH, INSPECTION, Keyless Entry System.>
	4. Check the output signal to the body integrated unit.	<Ref. to SL-19, CHECK OUTPUT SIGNAL TO BODY INTEGRATED UNIT, INSPECTION, Keyless Entry System.>
	5. Replace the keyless entry control module.	<Ref. to SL-55, Keyless Entry Control Module.>
The panic alarm does not operate.	1. Check the keyless transmitter battery and function.	<Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>
	2. Check the horn operation.	<Ref. to SL-19, CHECK HORN OPERATION, INSPECTION, Keyless Entry System.>
	3. Replace the keyless entry control module.	<Ref. to SL-55, Keyless Entry Control Module.>

Symptom	Repair order		Reference
Buzzer and hazard light do not operate.	1. Check the buzzer function.		<Ref. to SL-16, CHECK BUZZER SOUND SETTING, INSPECTION, Keyless Entry System.>
	2. Check the buzzer and hazard light operation.	Buzzer	<Ref. to SL-20, CHECK KEYLESS BUZZER, INSPECTION, Keyless Entry System.>
		Hazard light	<Ref. to SL-20, CHECK HAZARD LIGHT OPERATION, INSPECTION, Keyless Entry System.>
	3. Replace the keyless entry control module.		<Ref. to SL-55, Keyless Entry Control Module.>
Room light does not operate.	1. Check the room light operation.		<Ref. to SL-19, CHECK ROOM LIGHT OPERATION, INSPECTION, Keyless Entry System.>
	2. Replace the keyless entry control module.		<Ref. to SL-55, Keyless Entry Control Module.>
The door warning does not operate.	1. Check the door switch.		<Ref. to SL-17, CHECK DOOR SWITCH, INSPECTION, Keyless Entry System.>
	2. Check the buzzer operation.		<Ref. to SL-20, CHECK KEYLESS BUZZER, INSPECTION, Keyless Entry System.>
	3. Replace the keyless entry control module.		<Ref. to SL-55, Keyless Entry Control Module.>

2. CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION

	Step	Check	Yes	No
1	**CHECK KEYLESS TRANSMITTER BATTERY.** 1) Remove the battery from the transmitter. <Ref. to SL-57, REMOVAL, Transmitter.> 2) Check the battery voltage. <Ref. to SL-57, INSPECTION, Transmitter.>	Is the voltage 2.5 V or more?	Go to step **2**.	Replace the transmitter battery. <Ref. to SL-57, Transmitter.>
2	**CHECK KEYLESS TRANSMITTER.** Register the keyless transmitter which operates normally on other vehicles to the inspection target vehicle. <Ref. to SL-57, REGISTRATION OF KEYLESS TRANSMITTER WITH SUBARU SELECT MONITOR, REPLACEMENT, Transmitter.> 1) Close all the doors and rear gate of inspection target vehicle. 2) Using the keyless transmitter, lock and unlock the doors and rear gate of vehicle.	Is the inspection target vehicle operates lock and unlock normally?	Go to step **3**.	Due to vehicle malfunction, continue the keyless entry system diagnosis.
3	**CHECK KEYLESS TRANSMITTER.** Register the keyless transmitter of inspection target vehicle to the another vehicle which the keyless system operates normally. <Ref. to SL-57, REGISTRATION OF KEYLESS TRANSMITTER WITH SUBARU SELECT MONITOR, REPLACEMENT, Transmitter.>	Is the keyless transmitter registered correctly?	Go to step **4**.	Replace the keyless transmitter. <Ref. to SL-57, REGISTRATION OF KEYLESS TRANSMITTER WITH SUBARU SELECT MONITOR, REPLACEMENT, Transmitter.>
4	**CHECK KEYLESS TRANSMITTER.** Check the registered keyless transmitter. 1) Close all the doors and rear gate of the vehicle which operates keyless system normally. 2) Using the keyless transmitter, lock and unlock the doors and rear gate of vehicle.	Does the vehicle operates lock and unlock normally?	The transmitter is OK.	Replace the keyless transmitter. <Ref. to SL-57, REGISTRATION OF KEYLESS TRANSMITTER WITH SUBARU SELECT MONITOR, REPLACEMENT, Transmitter.>

CAUTION:
Be sure to reset the keyless transmitter of other vehicle which is registered to the inspection target vehicle, and the vehicle to which the keyless transmitter is registered for inspection, to the condition before performing the inspection. (Register the keyless transmitter again.)

3. CHECK BUZZER SOUND SETTING

	Step	Check	Yes	No
1	**CHECK BUZZER SOUND SETTING.** 1) Check the current setting for the buzzer. 2) Remove the key from ignition switch. 3) Close all the doors and the rear gate. 4) Press the LOCK/ARM or UNLOCK/DISARM button.	Does the buzzer sound?	The buzzer function is normal.	Go to step **2**.
2	**CHECK BUZZER SOUND SETTING.** 1) Open the driver's side door and pull the key out from the ignition switch. 2) With the vehicle side manual UNLOCK switch ON, insert the key in the ignition switch. 3) Within 10 seconds from 2), repeatedly pull out and insert the key 5 times. 4) Open and close the door within 5 seconds from the fifth time in 3). 5) The answer back buzzer setting is switched (ON ←→ OFF) and the hazard light flashes 3 times. NOTE: If the door is not opened and closed within 10 seconds, the hazard light will flash once. In this case, start over from the beginning. 6) Press the LOCK/ARM or UNLOCK/DISARM button.	Does the buzzer sound?	The buzzer function is normal.	Check the function of the transmitter. <Ref. to SL-15, CHECK KEYLESS TRANSMITTER BATTERY AND FUNCTION, INSPECTION, Keyless Entry System.>

4. CHECK FUSE

	Step	Check	Yes	No
1	**CHECK FUSE.** Remove and visually check the fuses No. 6 (in the main fuse box) and No. 3 (in the fuse & relay box).	Is the fuse blown out?	Replace the fuse with a new part.	Check the power supply and ground circuit. <Ref. to SL-16, CHECK POWER SUPPLY AND GROUND CIRCUIT, INSPECTION, Keyless Entry System.>

5. CHECK POWER SUPPLY AND GROUND CIRCUIT

	Step	Check	Yes	No
1	**CHECK POWER SUPPLY.** 1) Disconnect the keyless entry control module harness connector. 2) Measure the voltage between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 2 (+) — Chassis ground (–):*** ***(B176) No. 26 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **2**.	Check the harness for open circuits and shorts between the keyless entry control module and fuse.
2	**CHECK GROUND CIRCUIT.** Measure the resistance between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 14 — Chassis ground:***	Is the resistance less than 10 Ω?	The power supply and ground circuit are OK.	Repair the harness.

6. CHECK IGNITION SWITCH CIRCUIT

	Step	Check	Yes	No
1	**CHECK IGNITION SWITCH CIRCUIT.** 1) Disconnect the harness connector of the keyless entry control module. 2) Turn the ignition switch to ON. 3) Measure the voltage between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 10 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	The function of the ignition switch is normal.	Check the harness for open circuits or shorts between the keyless entry control module and the ignition relay.

7. CHECK DOOR SWITCH

	Step	Check	Yes	No
1	**CHECK DOOR SWITCH CIRCUIT.** Measure the voltage between the keyless entry control module harness connector terminal and the body ground. NOTE: If there is electrical noise, it may not be at 0 V. ***Connector & terminal*** ***Front, rear door and rear gate (wagon model):*** ***(B176) No. 6 (+) — Chassis ground (–):*** ***Trunk lid (sedan model):*** ***(B176) No. 7 (+) — Chassis ground (–):***	Is the voltage 0 V when any door or rear gate is opened?	Go to step **2**.	Go to step **3**.
2	**CHECK DOOR SWITCH CIRCUIT.** Measure the voltage between the keyless entry control module harness connector terminal and the body ground. ***Connector & terminal*** ***Front, rear door and rear gate (wagon model):*** ***(B176) No. 6 (+) — Chassis ground (–):*** ***Trunk lid (sedan model):*** ***(B176) No. 7 (+) — Chassis ground (–):***	Is the voltage 10 V or more when any door or rear gate is closed?	The door switch is OK.	Go to step **3**.
3	**CHECK DOOR SWITCH.** 1) Disconnect the door switch harness connector. 2) Measure the resistance between door switch terminals. ***Terminals*** ***Door switch No. 1 — No. 3:*** ***Rear gate latch switch (wagon model) No. 1 — No. 2:*** ***Trunk room light switch (sedan model) No. 1 — No. 2:***	Is the resistance 1 MΩ or more when door switch is pushed?	Go to step **4**.	Replace the door switch.
4	**CHECK DOOR SWITCH.** Measure the resistance between door switch terminals. ***Terminals*** ***Door switch No. 1 — No. 3:*** ***Rear gate latch switch (wagon model) No. 1 — No. 2:*** ***Trunk room light switch (sedan model) No. 1 — No. 2:***	Is the resistance less than 1 Ω when door switch is released?	Check the harness for open circuits and shorts between the keyless entry control module and door switch.	Replace the door switch.

8. CHECK KEY WARNING SWITCH

	Step	Check	Yes	No
1	**CHECK FUSE.** Remove and visually check the fuse No. 6 (in the main fuse box).	Is the fuse blown out?	Replace the fuse with a new part.	Go to step **2**.
2	**CHECK KEY WARNING SWITCH CIRCUIT.** 1) Disconnect the keyless entry control module harness connector. 2) Insert the key into ignition switch. (LOCK position) 3) Measure the voltage between harness connector terminal and chassis ground. ***Connector & terminal*** ***(B176) No. 9 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **3**.	Go to step **4**.
3	**CHECK KEY WARNING SWITCH CIRCUIT.** 1) Remove the key from ignition switch. 2) Measure the voltage between harness connector terminal and chassis ground. NOTE: If there is electrical noise, it may not be at 0 V. ***Connector & terminal*** ***(B176) No. 9 (+) — Chassis ground (–):***	Is the voltage 0 V?	The key warning switch is OK.	Go to step **4**.
4	**CHECK KEY WARNING SWITCH.** 1) Disconnect the key warning switch harness connector. 2) Insert the key into ignition switch. (LOCK position) 3) Measure the resistance between key warning switch terminals. ***Terminals*** ***No. 1 — No. 2:***	Is the resistance less than 1 Ω?	Go to step **5**.	Replace the key warning switch.
5	**CHECK KEY WARNING SWITCH.** 1) Remove the key from ignition switch. 2) Measure the resistance between key warning switch terminals. ***Terminals*** ***No. 1 — No. 2:***	Is the resistance 1 MΩ or more?	Check the following: • Harness for open circuits and shorts between the key warning switch and fuse • Harness for open circuits and shorts between the keyless entry control module and the key warning switch.	Replace the key warning switch.

9. CHECK ROOM LIGHT OPERATION

	Step	Check	Yes	No
1	**CHECK ROOM LIGHT OPERATION.** Make sure the room light illuminates when the room light switch is turned to ON.	Does the room light illuminate?	Go to step **2**.	Check the room light circuit.
2	**CHECK HARNESS BETWEEN ROOM LIGHT AND BODY INTEGRATED UNIT.** 1) Disconnect the body integrated unit harness connector and room light harness connector. 2) Measure the resistance between the body integrated unit harness connector terminal and room light harness connector terminal. ***Connector & terminal*** ***(B280) No. 5 — (R52) No. 2:***	Is the resistance less than 10 Ω?	The room light operation circuit is OK.	Check the harness for open or short between body integrated unit and room light.

10.CHECK OUTPUT SIGNAL TO BODY INTEGRATED UNIT

	Step	Check	Yes	No
1	**CHECK OUTPUT SIGNAL.** Measure the voltage between the keyless entry control module harness connector terminal and the chassis ground when the UNLOCK/DISARM button of the transmitter is pressed. ***Connector & terminal*** ***(B176) No. 18 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **2**.	Replace the keyless entry control module.
2	**CHECK OUTPUT SIGNAL.** Measure the voltage between the keyless entry control module harness connector terminal and the chassis ground when the LOCK/ARM button of the transmitter is pressed. ***Connector & terminal*** ***(B176) No. 19 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **3**.	Replace the keyless entry control module.
3	**CHECK HARNESS BETWEEN KEYLESS ENTRY CONTROL MODULE AND BODY INTEGRATED UNIT.** 1) Disconnect the keyless entry control module harness connector and body integrated unit harness connector. 2) Measure the resistance between the keyless entry control module harness connector and body integrated unit harness connector terminals. ***Connector & terminal*** ***(B176) No. 18 — (B281) No. 14:*** ***(B176) No. 19 — (B281) No. 13:***	Is the resistance less than 10 Ω?	Replace the body integrated unit.	Check the harness for open or short circuits between the keyless entry control module and body integrated unit.

11.CHECK HORN OPERATION

	Step	Check	Yes	No
1	**CHECK HORN OPERATION.** Make sure that the horn sounds when the horn switch is pushed.	Does the horn sound?	Go to step **2**.	Check the horn circuit.
2	**CHECK HORN OPERATION.** 1) Disconnect the harness connector of the keyless entry control module. 2) Connect the harness connector terminal to ground using a suitable lead wire. ***Connector & terminal*** ***(B176) No. 24 — Chassis ground:***	Does the horn sound?	Replace the keyless entry control module.	Check the harness for open circuits or shorts between the keyless entry control module and the horn relay.

12.CHECK HAZARD LIGHT OPERATION

	Step	Check	Yes	No
1	**CHECK HAZARD LIGHT OPERATION.** Make sure that the hazard light blinks when hazard switch is turned ON.	Does the hazard light blink?	Go to step **2**.	Check the hazard light circuit.
2	**CHECK OUTPUT SIGNAL.** 1) Remove the key from ignition switch. 2) Close all the doors and the rear gate. 3) Measure the voltage between the keyless entry control module harness connector terminal and the chassis ground when the LOCK/ARM button of the transmitter is pressed. ***Connector & terminal*** ***(B176) No. 1 (+) — Chassis ground (–):*** ***(B176) No. 13 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Check the harness for open circuits or shorts between the keyless entry control module and the turn signal lights.	Replace the keyless entry control module.

13.CHECK KEYLESS BUZZER

	Step	Check	Yes	No
1	**CHECK FUSE.** Remove and check fuse No. 2 (in the main fuse box).	Is the fuse blown out?	Replace the fuse with a new part.	Go to step **2**.
2	**CHECK POWER SUPPLY TO THE KEYLESS BUZZER.** 1) Disconnect the connector from the keyless buzzer. 2) Measure the voltage between the keyless buzzer harness connector and the chassis ground. ***Connector & terminal*** ***(F102) No. 2 (+) — Chassis ground (–):***	Is the voltage 10 V or more?	Go to step **3**.	Check for open or short circuit in the harness between the fuse and keyless buzzer.
3	**CHECK HARNESS BETWEEN KEYLESS BUZZER AND KEYLESS ENTRY CONTROL MODULE.** 1) Disconnect the connector from the keyless entry control module. 2) Measure the resistance in the harness between the keyless buzzer and keyless entry control module. ***Connector & terminal*** ***(F102) No. 1 — (B176) No. 3:***	Is the resistance less than 10 Ω?	Go to step **4**.	Repair the harness between the keyless buzzer and keyless entry control module.
4	**CHECK KEYLESS BUZZER.** Connect the battery positive (+) lead to terminal No. 2 of the keyless buzzer connector, and the negative (–) lead to terminal No. 1, and check whether or not the buzzer will sound.	Does the buzzer sound?	Replace the keyless entry control module.	Replace the keyless buzzer.